

ABC EXPRESS

Rencana Implementasi Keseluruhan Ontology Architecture, Data Warehouse, dan Operational Core ABC Express

Versi 1.0 - Read-only assessment berdasarkan Ontology Master, GitHub repositories, dan Supabase projects

Tanggal: 5 Juni 2026

Scope dokumen

Dokumen ini menyajikan rencana implementasi keseluruhan secara terstruktur. Analisis repo GitHub dan Supabase dilakukan secara read-only; tidak ada perubahan, migrasi, query mutasi, atau edit repository/database yang dilakukan.

Sistem	Repository	Supabase project/database	Peran target
ABC Talk	abc-express/abc-talk	ABC Talk (Production)	Conversation, Activity, Signal source
ANTERO	abc-express/antero-platform	Antero	Operational Ontology Core awal
OBL	abc-express/abc-obl	OBL Application	Performance consumer dan scorecard registry
CMS	abc-express/abc-cms	ABC CMS (Production)	Content/marketing source

Prepared for internal architecture planning. Do not distribute externally without ABC Express approval.

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1. Ringkasan Eksekutif

Kesimpulan utama

ABC Express tidak perlu membuat operational core baru dari nol. ANTERO sudah menjadi pusat gravitasi operasional dan sebaiknya dinaikkan secara bertahap menjadi Operational Ontology Core. Data warehouse tetap dibangun sebagai analytical SSOT, tetapi write-path operasional harus melalui API/action layer yang audited, bukan langsung ke warehouse.

Rencana ini menyatukan tiga kebutuhan: single source of truth untuk analitik dan audit, canonical object layer untuk operasional, serta ontology/action layer untuk aplikasi dan AI Agent. Fokus implementasi adalah mengurangi konflik entity antar database tanpa melakukan big-bang migration.

1.1 Target akhir

- Warehouse menjadi analytical single source of truth untuk reporting, TP, OBL, audit, BI, dan AI analysis.
- Antero menjadi operational system of record yang diperkuat dengan canonical IDs, MDM, source mapping, audit trail, dan action APIs.
- ABC Talk menjadi source utama Conversation, Activity, Signal, dan contact-channel identity.
- OBL menjadi consumer dari metric resmi dan pemilik scorecard/performance domain.
- CMS menjadi content/marketing source, bukan master untuk service area, SLA, route, atau operational coverage.
- Semua perubahan operasional lintas sistem berjalan melalui audited actions, outbox events, dan data contracts.

1.2 Keputusan arsitektur utama

Keputusan	Implikasi implementasi
Antero sebagai Operational Ontology Core awal	Tambah canonical layer dan action API di sekitar Antero; jangan bangun DB operasional kelima dulu.
Warehouse bukan write-path operasional	Warehouse menerima CDC/event/batch; aplikasi menulis lewat Ontology/Action API.
Canonical Location dan ServiceAreaCoverage sebagai pilot	Menyelesaikan konflik kota/area antara Antero, ABC Talk, CMS, dan kebutuhan pricing/SLA.
Source mapping wajib untuk semua entity lintas sistem	Setiap ID dari source tetap disimpan, tetapi diikat ke canonical ID.
MDM ringan sebelum MDM enterprise	Bangun entity registry, source mapping, conflict queue, survivorship rules, dan steward workflow.
AI Agent hanya boleh action via catalog	Tidak boleh direct write ke database produksi; semua action versioned, permissioned, dan audited.

1.3 Minimum viable platform

MVP architecture sketch

```
Supabase source projects
- Antero
- ABC Talk (Production)
- OBL Application
- ABC CMS (Production)
  |
  | CDC / batch / API / event feed
  v
AWS S3 Raw Landing + Redshift + dbt
  |
  v
Canonical models + semantic metrics
  |
  +--> OBL official metrics
  +--> BI / audit / finance / TP
  +--> Ontology API read layer
  +--> AI Agent context layer
```

Operational writes:
App / AI Agent -> Action API -> Antero/Owning App -> Outbox -> Warehouse

2. Prinsip dan Decision Locks

2.1 Prinsip desain

1. Object-first, not table-first: model bisnis mengikuti entity ontology seperti Account, Order, Shipment, Trip, Manifest, ShipmentLeg, Dispatch, Activity, Signal, ServiceAreaCoverage.
2. Source of record dan source of truth dibedakan: database aplikasi tetap dapat menjadi source of record untuk transaksi, sedangkan canonical/ontology layer menyatakan truth lintas sistem.
3. No big-bang migration: integrasi dimulai read-only, lalu canonical read, lalu canonical write secara bertahap.
4. Lineage tidak boleh hilang: setiap source PK harus dipetakan ke canonical ID dan tetap dapat diaudit.
5. Operational write harus audited: perubahan status, assignment, coverage, SLA, customer tier, atau routing harus punya actor, timestamp, reason, dan before/after state.
6. Warehouse tidak menjadi operational database: warehouse optimal untuk query analitik, bukan transaksi operasional dan workflow.
7. Action Catalog menjadi kontrak resmi untuk AI dan aplikasi: AI Agent hanya memanggil action yang eksplisit, permissioned, dan logged.

2.2 Decision locks untuk fase 1

Decision lock	Status	Rationale
DL-01: Antero sebagai operational core awal	Locked for MVP	Antero sudah memegang POS/resi, manifest, dispatch, tracking, finance, invoice, vendor, master data, TP, trip.
DL-02: Canonical Location/ServiceAreaCoverage sebagai pilot	Locked for MVP	Konflik kota/area paling mudah dikontrol dan berdampak ke pricing, SLA, routing, dan coverage claim.
DL-03: ABC Talk bukan Account master	Locked	Contacts di ABC Talk adalah WhatsApp/contact-channel identity, bukan legal Account.
DL-04: OBL mengambil metric resmi dari semantic layer	Locked	Menghindari definisi KPI bercabang antar aplikasi.
DL-05: CMS bukan operational coverage source	Locked	Content area/campaign hanya untuk marketing; operational coverage harus dari ServiceAreaCoverage.
DL-06: Write-back AI tidak direct-to-DB	Locked	Semua AI actions harus melewati API/action catalog dan audit log.

2.3 Non-goals fase 1

- Tidak melakukan rename besar-besaran tabel produksi.
- Tidak memindahkan semua aplikasi ke satu database baru.
- Tidak menutup database existing atau memutus vendor feed.
- Tidak membuat MDM enterprise berat sebelum canonical pilot berhasil.
- Tidak menjadikan abc-ontology/DAEMON sebagai mandatory runtime dependency produksi sebelum proof-of-fit.

3. Temuan Current State: Repo dan Database

3.1 Supabase estate

Project/database	Region/status	Peran saat ini	Catatan implementasi
Antero	ap-southeast-1 / active healthy	Core operasional logistik	Kandidat operational ontology core.
ABC Talk (Production)	ap-southeast-1 / active healthy	WhatsApp CRM/conversation	Signal, Activity, Contact-channel source.
OBL Application	ap-southeast-1 / active healthy	Scorecard/performance app	Consumer metric resmi; owner scorecard config.
ABC CMS (Production)	ap-southeast-1 / active healthy	Strapi CMS	Content source; bukan operational truth.
ABC CMS (Staging)	ap-southeast-1 / active healthy	Staging CMS	Useful untuk release workflow dan testing content.

3.2 Antero - operational core kandidat

- Repository: abc-express/antero-platform.
- Framework: Next.js 16, React 19, TypeScript, Mantine UI, Tailwind CSS, TanStack React Query, Supabase, direct PostgreSQL via pg.Pool.
- Functional scope: POS/resi, manifest, dispatch, tracking, pickup, finance, invoice, vendor, master data, RBAC, logs.
- Database highlights: customer 47,039; pos 73,822; pos_detail 84,723; pos_tracking 163,614; manifest 13,364; manifest_detail 38,075; dispatch 2,881; dispatch_detail 10,768; routeprice 12,755; vendor 331; tp_attributions 10,227; trip 13,364; routing_decision 0.

Interpretasi

Antero sudah menjadi sistem operasional paling kaya. Namun canonical truth tidak boleh diasumsikan otomatis: geografi, customer, trip/manifest boundary, dan shipment-leg model perlu dinormalisasi dan diberi canonical mapping.

3.3 ABC Talk - conversation, activity, signal source

- Repository: abc-express/abc-talk.
- Framework: Next.js App Router, React 19, TypeScript strict, MUI, Supabase auth/database/realtime, Vercel.
- Functional scope: unified WhatsApp inbox, team management, conversation assignment, ownership transfer, templates, realtime updates, push notification, export.
- Database highlights: contacts 821; conversations 788; messages 16,631; conversation_events 10,142; webhook_logs_raw 34,649; webhook_events_normalized 33,305; bot_tariffs 7,631; location_resolutions 194; integration_webhook_events 2,830; sync_changes 69,942.

Interpretasi

ABC Talk tidak menjadi master Account. Ia menjadi sumber event untuk Conversation, Activity, Signal, Lead, dan Contact. Mapping ke canonical Account harus explicit dan nullable sampai matching selesai.

3.4 OBL - performance domain

- Repository: abc-express/abc-obl.
- Framework: Next.js 14, React 18, Supabase, Recharts, Zustand.
- Functional scope: OBL Scorecard, KPI, behavior, leverage, review, role access, push subscriptions.
- Database highlights: organizations 13; users 76; scorecards 105; kpi_entries 587; behavior_scores 525; leverage_scores 315; kpi_templates 294; ro_kpi_profiles 24; quarterly_reviews 3; kpi_rollup_mappings 123.

3.5 CMS - content domain

- Repository: abc-express/abc-cms.
- Framework: Strapi 5.45.1, PostgreSQL via pg, users-permissions, documentation plugin, Supabase bucket upload provider.
- Database highlights: blog_posts 226; blog_categories 12; campaigns 428; areas 114; files 401; Strapi admin/config tables.

Interpretasi

CMS areas harus diperlakukan sebagai marketing/content taxonomy. Service coverage resmi tetap harus berasal dari canonical ServiceAreaCoverage.

3.6 abc-ontology repo

- Repository: abc-express/abc-ontology.
- Deskripsi: generic ontology platform monorepo dengan bounded contexts collect-sensing, ontology, read-write-loops, action-runtime, API, policy/security.
- Runtime map: Go, Rust, TypeScript/NestJS, Python SDK potential.
- Rekomendasi: jadikan reference architecture dan sandbox jangka menengah; jangan dijadikan mandatory production dependency untuk MVP sebelum proof-of-fit.

4. Arsitektur Target

4.1 Layer arsitektur

```
Layer 0 - Source Systems
  Antero DB | ABC Talk DB | OBL DB | CMS DB | Vendor/External DB/API

Layer 1 - Collection and Landing
  CDC / batch / API extract / webhook event capture / outbox relays

Layer 2 - Raw and Staging Warehouse
  S3 raw landing -> Redshift raw/stg -> dbt tests and standardization

Layer 3 - Canonical and MDM
  canonical object tables, source mappings, conflict queue, survivorship rules

Layer 4 - Semantic and Metrics
  official metrics: SLA, NLM, OBL KPI inputs, revenue, cost, coverage, signals

Layer 5 - Operational Ontology API
  read layer, action layer, permission layer, audit trail, outbox events

Layer 6 - Consumption
  ANTERO modules, ABC Talk, OBL, CMS, BI, AI Agent, audit, finance
```

4.2 Target deployment topology

Component	Recommended stack	Notes
Operational DB	Supabase/Postgres for Antero; optional Aurora later	Keep existing Supabase projects; avoid disruptive migration.
Data landing	Amazon S3	Partition by source_system/table/date.
Warehouse	Amazon Redshift / Redshift Serverless	Central analytical SSOT.
Transform	dbt	Staging, canonical, marts, tests, docs.
Orchestration	Dagster or AWS Step Functions/MWAA	Dagster better for data asset lineage; Step Functions simpler AWS-native.
CDC/extract	AWS DMS / Airbyte / custom API jobs	Start pragmatic; event/outbox for near-

Component	Recommended stack	Notes
		real-time operational actions.
Event bus	EventBridge + SQS/SNS	Kafka/MSK only if event scale/complexity justifies.
Metadata catalog	OpenMetadata or DataHub	Glossary, ownership, lineage, data quality.
Action/API layer	Next.js API/NestJS/Fastify	Versioned OpenAPI; audited.
Auth/policy	Supabase Auth now; centralized IAM/SSO later	Policy hardening needed before AI write-back.

4.3 Operational write pattern

Good pattern:
 User/App/AI -> Action API -> validate ontology -> permission check
 -> write owning system -> audit_events -> outbox_events
 -> warehouse/semantic refresh -> notifications

Avoid:
 App A writes DB A
 App B writes DB B
 Warehouse later tries to guess which row is true

4.4 Read pattern

Operational read:
 App -> Ontology read API -> canonical object + source lineage

Analytical read:
 BI/OBL/Finance/AI analysis -> semantic layer -> warehouse marts

Legacy read:
 Legacy app -> existing DB until replaced, but master data reads migrate first

5. Object Ownership Matrix

Ownership matrix menetapkan siapa yang menjadi source of record sekarang, siapa canonical owner target, dan bagaimana konflik diselesaikan. Matrix ini wajib menjadi dokumen hidup yang di-review minimal bulanan pada fase implementasi.

Ontology object	Current source terkuat	Canonical owner target	Catatan
Account	Antero.customer; ABC Talk contacts as weak signal	Antero/Ontology Core	Account = legal customer entity; tidak sama dengan WhatsApp contact.
Contact	ABC Talk.contacts; Antero.contact	Ontology Core	Perlu mapping wa_id/email/phone ke canonical_contact_id.
Conversation	ABC Talk.conversations/messages/events	ABC Talk	Push-derived Activity dan Signal ke canonical layer.
Activity	ABC Talk events; Antero shipment events; finance events	Ontology Core/warehouse	Activity adalah atomic touchpoint; dibentuk dari events.

Ontology object	Current source terkuat	Canonical owner target	Catatan
Signal	ABC Talk sentiment/AI; Antero ops; OBL metrics	Ontology Core	Unified signal table lintas app.
Order	Antero order/POS-related flow	Antero/Ontology Core	Perlu eksplisitkan Order -> Shipment -> TTK boundary.
Shipment	Antero.pos/pos_detail/tracking	Antero/Ontology Core	Map POS/resi ke Shipment/TTK sesuai ontology.
TTK/Resi	Antero POS/AWB/resi flow	Antero/Ontology Core	Human reference utama untuk tracking, POD, audit.
Trip	Antero.trip	Antero/Ontology Core	Cek apakah Trip sudah 1:N Manifest atau masih mirrored.
Manifest	Antero.manifest/manifest_detail	Antero/Ontology Core	Sudah mature; perlu canonical relation ke ShipmentLeg.
ShipmentLeg	Belum jelas sebagai explicit table	Ontology Core	Critical junction object: Shipment x Manifest.
Dispatch	Antero.dispatch/dispatch_detail	Antero/Ontology Core	Last-mile document; link ke TTK/POD.
RoutingDecision	Antero.routing_decision	Antero/Ontology Core	Table ada namun kosong; cocok untuk activation phase.
ServiceAreaCoverage	Antero city/kantor coverage; ABC Talk locations; CMS areas	Ontology Core	Pilot utama untuk MDM dan conflict resolution.
TP Calculation	Antero tp_* tables	Antero/TP + warehouse	Official metrics roll up ke finance/OBL.
OBL Scorecard	OBL DB	OBL	Operational KPI input dari semantic layer.
Content/Campaign	CMS DB	CMS	Tidak menjadi master operasional.

5.1 Ownership states

State	Meaning	Examples
AUTHORITATIVE	Source boleh menjadi dasar canonical truth untuk field tertentu.	Antero for manifest operational facts; OBL for scorecard config.
CONTRIBUTING	Source memberi sinyal, tetapi tidak menentukan truth sendiri.	ABC Talk contact name, CMS areas, user notes.
REFERENCE_ONLY	Source untuk display/context; tidak digunakan untuk overwrite canonical truth.	CMS campaign labels, public content tags.
LEGACY	Masih dibaca untuk continuity; write baru harus diarahkan ke API/core.	Old master data tables after migration.
VENDOR_FEED	Data eksternal dengan SLA/contract; perlu timestamp dan confidence.	Vendor DB/API, Accurate integrations, route/pricing sources.

6. Canonical Data, MDM, dan Conflict Resolution

6.1 Minimal canonical schema

```

core.entity_registry
  canonical_id, entity_type, human_id, canonical_name, status,
  lifecycle_state, valid_from, valid_to, created_at, updated_at

core.source_entity_mappings
  entity_type, source_system, source_database, source_schema,
  source_table, source_pk, canonical_id, match_method,
  confidence_score, first_seen_at, last_seen_at

core.entity_conflicts
  conflict_id, entity_type, canonical_id, field_name,
  source_a, value_a, source_b, value_b, suggested_value,
  severity, steward_owner, resolution_status, resolved_by, resolved_at

core.survivorship_rules
  entity_type, field_name, priority_order, fallback_rule,
  manual_review_threshold, effective_from, effective_to

core.audit_events
  actor_type, actor_id, action_code, entity_type, entity_id,
  before_state, after_state, reason_code, request_id, created_at

core.outbox_events
  event_type, aggregate_type, aggregate_id, payload, status,
  retry_count, created_at, published_at

```

6.2 Location dan ServiceAreaCoverage sebagai pilot

Source	Relevant tables/fields	Treatment
Antero	city, province, provinsi, kantor, kantor_coverage_kota, routeprice	Primary operational source, but normalize codes/names.
ABC Talk	origin, destination, origin_locations, destination_locations, location_resolutions, bot_tariffs	Contributing source for fuzzy aliases and customer language.
CMS	areas	Marketing/content taxonomy; map to canonical location where applicable.
External reference	BPS/Kemendagri codes if available	Preferred authority for official kab/kota codes.

```

Example conflict:
Antero.city.name      = Jakarta Selatan
ABC Talk.destination  = Kota Adm. Jakarta Selatan
CMS.areas.name        = Jaksel

Canonical:
location_id           = LOC-3174
location_type          = kab_kota
canonical_name         = Kota Administrasi Jakarta Selatan
province_name         = DKI Jakarta
source aliases        = Jakarta Selatan, Kota Adm. Jakarta Selatan, Jaksel

```

6.3 Matching strategy

Entity	Phase 1 matching	Phase 2 matching	Manual review trigger
Location	Exact code; normalized name; province context	Alias dictionary; fuzzy match; geospatial reference	Confidence < 0.90 or multiple candidates.
Account	NPWP, legal name, customer	Phone/email/domain/address	Potential duplicate with

Entity	Phase 1 matching	Phase 2 matching	Manual review trigger
	code	similarity	revenue impact.
Contact	wa_id, phone, email	Name + account + conversation context	Contact maps to multiple accounts.
Shipment/TTK	TTK/AWB number, POS ID	Order/project context	Duplicate TTK or inconsistent status.
Vendor/Agent	Vendor ID/name/phone	Area/capability similarity	Tier 4/5 or financial exposure.

6.4 Stewardship workflow

1. Ingestion detects mismatch or duplicate candidates.
2. Rule engine creates entity_conflicts row.
3. Data steward reviews suggested resolution.
4. Steward approves/rejects/merges/splits canonical entity.
5. Action API writes canonical update and audit_event.
6. Source mapping is updated; downstream marts refresh.
7. If operational app needs change, write-back event is published.

7. Data Warehouse dan Semantic Layer

7.1 Warehouse zones

Zone	Purpose	Examples
raw	Salinan source apa adanya; immutable/append-oriented where possible.	raw_antero.pos, raw_abc_talk.messages, raw_obl.scorecards, raw_cms.campaigns.
staging	Type casting, naming standard, basic cleaning, dedupe, source metadata.	stg_antero_shipments, stg_talk_conversations, stg_obl_kpi_entries.
core/canonical	Official entity models dan source mapping.	core_location, core_account, core_shipment, core_manifest, core_signal.
mart	Domain-specific analytical views.	mart_operations_sla, mart_finance_nlm, mart_customer_health, mart_obl_inputs.
semantic	Official metric definitions and business glossary.	on_time_delivery_rate, manifest_density, NLM, active_coverage_count.

7.2 dbt model naming

```
models/
  sources/
    src_antero.yml
    src_abc_talk.yml
    src_obl.yml
    src_cms.yml
  staging/
    stg_antero__pos.sql
    stg_antero__manifest.sql
    stg_abc_talk__messages.sql
    stg_obl__scorecards.sql
    stg_cms__campaigns.sql
  core/
    core_locations.sql
    core_service_area_coverage.sql
    core_accounts.sql
```

```

core_contacts.sql
core_shipments.sql
core_manifests.sql
core_shipment_legs.sql
core_signals.sql
marts/
  mart_operations__shipment_sla.sql
  mart_finance__nlm.sql
  mart_customer__health_score.sql
  mart_obl__scorecard_inputs.sql

```

7.3 Data quality gates

Gate	Rule examples	Blocking?
Source freshness	Antero POS updated within SLA; ABC Talk messages event lag below threshold.	Warn first, block for critical marts.
Primary key uniqueness	TTK number unique; canonical_id unique per entity_type.	Block canonical publish.
Referential integrity	Shipment.destination_location_id exists in core_locations.	Block canonical publish.
Status enum validity	Shipment status, coverage_status, signal severity in approved enums.	Block.
Financial consistency	Sum manifest allocation equals trip cost within tolerance.	Block finance/TP mart.
PII classification	Contact data tagged and excluded from broad BI marts.	Block if unclassified.

7.4 Semantic layer metrics

Metric	Owner	Primary source	Consumer
on_time_delivery_rate	COO / Ops Data Steward	Antero tracking + core_shipment_legs	OBL, Ops dashboard, Account health.
manifest_density	Operations Planner	Antero manifest_detail + capacity reference	OBL, route optimization.
NLM per shipment/account/RO	CFO / TP owner	Antero tp_* + vendor cost allocation	Finance, OBL, Account Plan.
active_coverage_count	COO / Coverage steward	core_service_area_coverage	Marketing claim, IPO data room.
customer_health_score	CMO / Customer Intelligence	Shipments, SLA, invoices, activities	AM dashboard, signals.
lead_response_sla	CMO / ABC Talk owner	ABC Talk conversations/messages	Sales/CS dashboard, signals.

8. Operational Action Layer dan AI Agent

8.1 Action pattern

Action request lifecycle:

1. request received with actor, action_code, input payload
2. schema validation against ontology spec
3. authorization/policy check
4. idempotency key check
5. business rule validation

6. transaction against owning system
7. audit_events insert
8. outbox_events insert
9. response with action_id and resulting canonical object

8.2 Initial Action Catalog

Action	Owner system	Risk level	Notes
resolve_location_conflict	Ontology Core	Medium	Human steward approval required.
link_contact_to_account	Ontology Core + ABC Talk	Medium	PII handling and confidence threshold.
create_signal	Ontology Core	Low/Medium	Rule engine or AI may propose; owner acknowledges.
update_signal_status	Ontology Core	Low	Ack/action/resolve workflow.
update_service_area_coverage	Ontology Core/Antero	High	Affects order acceptance, SLA, marketing claim.
create_routing_decision	Antero	Medium	Start as manual/audit record.
update_shipment_status	Antero	High	Requires role, reason, idempotency, audit.
create_manifest	Antero	High	Warehouse/gudang role; validates trip/shipments.
create_shipment_leg	Ontology Core/Antero	High	Critical junction for allocation and SLA.
publish_obl_metric_snapshot	Warehouse/OBL	Medium	No manual overwrite of source facts.

8.3 API boundaries

```

Read APIs:
GET /ontology/locations
GET /ontology/service-areas
GET /ontology/accounts/{id}
GET /ontology/shipments/{id}
GET /ontology/ttk/{number}
GET /ontology/signals

Action APIs:
POST /actions/resolve-location-conflict
POST /actions/link-contact-to-account
POST /actions/create-signal
POST /actions/update-service-area-coverage
POST /actions/create-routing-decision
POST /actions/update-shipment-status
POST /actions/create-shipment-leg

```

8.4 AI Agent policy

- AI Agent boleh membaca warehouse/semantic layer dan ontology read API sesuai permission.
- AI Agent tidak boleh direct SQL ke production database.
- AI Agent hanya boleh melakukan write melalui Action Catalog.
- High-risk actions harus require human approval atau staged workflow.
- Setiap action harus punya idempotency key, request_id, audit trail, before/after state, dan owner.

- AI-generated suggestions harus disimpan sebagai proposal, bukan final truth, sampai rule atau steward menyetujui.

9. Security, Governance, dan Auditability

9.1 Immediate security observations

Area	Observation	Recommended action
ABC Talk	sync_changes RLS disabled; security advisor flags critical exposure risk.	Design RLS policies before enabling; restrict REST exposure; review use by app/realtime.
ABC Talk	Several functions mutable search_path; some permissive insert policies; some SECURITY DEFINER callable by anon.	Harden functions with explicit search_path; revoke public execute; tighten policies.
ABC CMS	Many Strapi tables have RLS disabled.	Likely normal for server-side Strapi, but ensure Supabase REST exposure is not public. Do not enable RLS blindly.
Antero	Many tables RLS enabled but no policy advisory.	Audit whether app uses direct server DB or browser Supabase. Add explicit policy model where client access exists.
All systems	Multiple apps have their own user/role concepts.	Create identity mapping and central permission vocabulary before cross-app actions.

Important

Do not auto-apply RLS remediation SQL in production without policy design and app regression testing. Enabling RLS without policies can block application access. The correct sequence is inventory -> policy design -> staging test -> monitoring -> production rollout.

9.2 Governance roles

Role	Responsibility
Executive Sponsor	Approves ontology governance, major changes, funding, and conflict escalation.
Data Platform Owner	Owns warehouse, ingestion, orchestration, data quality, monitoring.
Ontology Owner	Owns object definitions, canonical IDs, schema registry, action catalog.
Domain Data Steward - Operations	Owns shipment, manifest, trip, dispatch, service area resolution.
Domain Data Steward - Commercial	Owns account, contact, lead, conversation, signal semantics.
Domain Data Steward - Finance/TP	Owns NLM, vendor cost allocation, invoice alignment, TP calculations.
App Owners	Own respective source apps and data contracts.
Security Owner	Owns RLS, service keys, access policies, audit controls.

9.3 Change governance

Ontology change types:

Minor: description, examples, typo, non-breaking enum label

Standard: new field, new action, new mapping, new metric

Major: new object, cardinality change, status lifecycle change, deletion rule

Approval:

Minor -> Ontology Owner + domain steward

Standard -> Ontology Owner + app owner + affected domain owner

Major -> Executive Sponsor + CTO/CDO + domain heads

9.4 Audit events minimum payload

```
{
  "audit_event_id": "uuid",
  "request_id": "uuid",
  "actor_type": "user|system|ai_agent|integration",
  "actor_id": "...",
  "action_code": "update_service_area_coverage",
  "entity_type": "ServiceAreaCoverage",
  "entity_id": "SAC-3174",
  "before_state": {...},
  "after_state": {...},
  "reason_code": "coverage_revalidation",
  "approval_id": "optional",
  "source_system": "ontology_api",
  "created_at": "timestamp"
}
```

10. Roadmap Implementasi 180 Hari

10.1 Phase overview

Phase	Timeline	Theme	Main deliverables
0	Hari 0-15	Mobilization and governance	RACI, source registry, object ownership draft, security freeze rules.
1	Hari 16-45	Location and ServiceAreaCoverage pilot	Canonical locations, source mappings, conflict queue, read API.
2	Hari 31-75	Warehouse foundation	S3 landing, Redshift, dbt staging, initial DQ tests, source freshness.
3	Hari 61-105	Account/Contact/Conversation integration	Account/contact mapping, ABC Talk activity feed, signal seed.
4	Hari 91-135	Shipment/Manifest/Trip/ ShipmentLeg canonicalization	Canonical shipment and leg model, allocation-ready relations.
5	Hari 120-160	Semantic metrics and OBL integration	Official KPI inputs, OBL metric snapshot, finance/TP marts.
6	Hari 145-180	Action Catalog and controlled write-back	Audited actions, approvals, outbox, first AI-safe operational actions.

10.2 Phase 0 - Mobilization and governance

- Appoint temporary Ontology Owner and Data Platform Owner.
- Create source_system_registry for Antero, ABC Talk, OBL, CMS, vendor systems.
- Freeze direct schema expansion for canonical master objects unless reviewed.
- Define initial severity levels for conflicts and data quality failures.
- Run non-mutating security review of Supabase exposure, RLS, service role usage, and public REST schemas.

10.3 Phase 1 - Location and ServiceAreaCoverage pilot

- Extract Antero city/province/provinsi/kantor/kantor_coverage_kota and CMS areas as read-only sources.
- Create canonical location model and mapping table in staging/warehouse first.
- Build conflict report: duplicate names, missing province, code mismatch, active/inactive ambiguity.
- Define stewardship UI or spreadsheet workflow for resolution.
- Expose read-only API for canonical service areas before any write-back.

10.4 Phase 2 - Warehouse foundation

- Set up S3 landing buckets and naming standard.
- Ingest four Supabase projects via CDC/batch using read-only credentials.
- Build dbt sources and staging models for top operational tables.
- Implement tests for uniqueness, not-null, accepted values, freshness, referential integrity.
- Create first marts: operations shipment tracking, manifest density, ABC Talk conversation activity, OBL scorecard input readiness.

10.5 Phase 3 - Account/Contact/Conversation integration

- Map ABC Talk contacts to canonical contacts and, where possible, to Antero customer accounts.
- Create contact-account link confidence scoring.
- Materialize Conversation and Activity models from ABC Talk.
- Create seed signal rules: lead response breach, negative sentiment, unresolved conversation, customer quiet.
- Do not overwrite Antero customer master automatically; steward review required for merges.

10.6 Phase 4 - Shipment/Manifest/Trip/ShipmentLeg canonicalization

- Define exact mapping from Antero POS/resi tables to ontology Shipment and TTK.
- Validate manifest_detail as source for ShipmentLeg candidate rows.
- Inspect trip-to-manifest cardinality; fix canonical model even if current physical table has mirrored rows.
- Materialize canonical_shipment_legs with sequence, leg_type, origin_in_leg, destination_in_leg, SLA, allocated cost fields.
- Prepare Three-Step allocation validation tests.

10.7 Phase 5 - Semantic metrics and OBL integration

- Define official OBL metric input tables from warehouse.
- Map OBL organizations/users/job_positions to canonical RO/Employee where possible.
- Replace ad-hoc OBL operational metric pulls with semantic layer snapshots.
- Publish metric definitions and ownership in catalog.
- Create reconciliation dashboard: OBL scorecard input vs operational facts.

10.8 Phase 6 - Action Catalog and controlled write-back

- Implement low-risk actions first: create_signal, update_signal_status, resolve_location_conflict.
- Add high-risk actions behind approval: update_service_area_coverage, update_shipment_status, create_shipment_leg.
- Implement audit_events and outbox_events.
- Define AI Agent permission scopes and approval workflow.
- Run shadow mode before allowing AI-assisted operational write-back.

11. Workstream, Backlog, dan Deliverables

Workstream	Owner	First 30-day deliverables	Dependencies
WS-A Ontology Spec	Ontology Owner	Object dictionary v0.1; enum registry; action catalog draft.	Ontology Master, app owner input.
WS-B Data Platform	Data Platform Owner	S3/Redshift/dbt skeleton; source configs; freshness checks.	AWS credentials, Supabase read-only access.
WS-C MDM Pilot	Operations Steward	Canonical location model; conflict queue; steward workflow.	Antero/CMS/ABC Talk extracts.
WS-D Antero Core	Antero App Owner	Object mapping from POS/Manifest/Dispatch/Trip to ontology.	Schema inspection and business validation.
WS-E ABC Talk Integration	ABC Talk App Owner	Conversation/Activity feed mapping; signal seed rules.	Contact-account matching rules.
WS-F OBL Metrics	OBL App Owner	Official metric input contract; OBL snapshot design.	Warehouse semantic layer.
WS-G Security	Security Owner	RLS exposure inventory; service key inventory; function hardening plan.	App owner validation.

11.1 Backlog by priority

Priority	Backlog item	Definition of done
P0	Source registry and data contracts	All production DBs and key tables have owner, refresh SLA, classification, and extraction method.
P0	Canonical Location pilot	At least Antero city/province and CMS areas mapped to canonical locations with conflict report.
P0	Security review before new integrations	No new integration uses service role in browser; RLS/public exposure risk documented.
P1	Warehouse raw/staging	Top 20 source tables ingested read-only and validated by dbt tests.
P1	Account/contact mapping	ABC Talk contacts linked to canonical contacts/accounts with confidence and review queue.
P1	Shipment ontology mapping	POS/resi/manifest/tracking tables mapped to Shipment/TTK/Manifest/Dispatch.
P2	ShipmentLeg materialization	Canonical leg table built and reconciled against manifest_detail.
P2	Semantic OBL metrics	OBL receives official snapshot; no ad-hoc conflicting metric definitions.
P2	Action Catalog v0.1	At least 5 safe actions implemented in shadow or controlled mode.

11.2 Data contracts minimum fields

```
source_system: antero
source_database: Antero
source_schema: public
source_table: manifest
owner: Operations / Antero App Owner
classification: operational_fact
refresh_sla: 15 minutes for operational marts, daily for audit snapshot
primary_key: id
natural_keys: manifest_number, manifest_id_human where available
breaking_change_policy: app owner must notify data platform before rename/drop/type change
pii_fields: none or list explicitly
known_issues: describe current data quality limitations
canonical_objects: Manifest, ShipmentLeg candidate
```

12. RACI dan Operating Model

Decision / activity	R	A	C	I
Ontology object definition	Ontology Owner	Executive Sponsor/CTO	Domain stewards, app owners	All product/data teams
Canonical Location resolution	Ops Data Steward	COO/Ontology Owner	Antero, CMS, ABC Talk owners	Commercial/Marketing
Warehouse pipeline	Data Platform Owner	CTO/CDO	App owners, Security	Domain owners
Security/RLS change	Security Owner	CTO	App owners, Data Platform	All affected users
Operational action approval	Domain Steward	Domain Head	Security, Ontology Owner	Auditors/BI/AI team
OBL metric definition	OBL Owner/Data Platform	COO/CHRO/CEO as applicable	Finance/Ops/Commercial	RO Heads
CMS content referencing operational data	CMS Owner	CMO	Ops Data Steward	Marketing/Sales

12.1 Operating cadence

Cadence	Meeting	Purpose	Output
Weekly	Implementation standup	Track blockers and sprint deliverables.	Decision/action log.
Weekly	Data quality review	Review freshness, failed tests, new conflicts.	DQ exceptions and owners.
Biweekly	Ontology review	Approve minor/standard object/action changes.	Versioned ontology spec.
Monthly	Stewardship council	Review conflict backlog, canonical ownership, security risk.	Resolved conflicts and policy decisions.
Quarterly	Architecture review	Reassess roadmap, scale, governance, tech debt.	Updated target architecture and budget asks.

13. Risk Register dan Mitigasi

Risk	Impact	Likelihood	Mitigation
Antero schema does not map cleanly to Ontology Master	High	Medium	Create canonical views first; avoid physical rewrite; validate with Ops.
Trip table mirrors Manifest instead of true 1:N Trip model	High	Medium	Materialize canonical Trip/Manifest relation in warehouse; fix write model later.
Location conflict causes wrong SLA/pricing/order acceptance	High	High	Make Location/ServiceAreaCoverage the first MDM pilot with steward approval.
ABC Talk contact mapped to wrong Account	Medium/High	Medium	Use confidence thresholds and manual review; never auto-merge high-value accounts.
Warehouse becomes de facto operational DB	High	Medium	Enforce write policy: operational writes only via Action API/owning app.
RLS hardening breaks production apps	High	Medium	Inventory access paths, test in staging, staged rollout, monitoring.
AI Agent writes unsafe changes	High	Medium	Action catalog, approvals, audit, idempotency, shadow mode.
Metric definitions diverge between OBL, Finance, Ops	Medium/High	High	Semantic layer with metric ownership and dbt tests.
Vendor/external DB feed is unreliable	Medium	Medium	Contract SLAs, freshness alerts, confidence flags, fallback process.
Team capacity bottleneck	High	Medium	Start with limited pilot; appoint stewards; avoid overbuilding platform.

13.1 Red flags to monitor weekly

- New production table added without source registry entry.
- Dashboard or OBL metric built directly from a source DB without semantic definition.
- AI/automation request requiring direct DB credential.
- Service area shown in CMS/marketing but missing/inactive in canonical coverage.
- Same customer/contact appears with multiple identifiers and no source mapping.
- TP/NLM calculation result differs between Antero and warehouse beyond tolerance.
- RLS/policy change deployed without staging verification.

14. Acceptance Criteria dan KPI Keberhasilan

14.1 Platform acceptance criteria

Area	Acceptance criteria
Source inventory	100% of four production DBs registered with owner, classification, extract method, and refresh SLA.

Area	Acceptance criteria
Canonical location	>=95% of operational city/area references mapped to canonical IDs; all low-confidence mappings queued for stewardship.
Warehouse	Top priority source tables landed, staged, and tested; freshness dashboard active.
MDM	source_entity_mappings and entity_conflicts live for Location and Contact/Account pilot.
Semantic layer	At least 10 official metrics documented with owner, formula, source, refresh cadence.
Action API	At least 5 actions implemented with validation, audit, idempotency, and permission checks.
Security	RLS/public exposure risks documented; no service role key in browser/client code; high-risk functions reviewed.

14.2 Business success metrics

Metric	Target after 180 days
Duplicate/conflicting city records	Known, queued, and reduced by >80% for priority locations.
Coverage claim accuracy	Marketing/BI coverage numbers generated only from canonical ServiceAreaCoverage.
Time to trace shipment truth	One canonical view by TTK/AWB including source lineage, tracking, manifest, dispatch, POD.
OBL metric consistency	OBL KPI inputs sourced from semantic layer, not conflicting ad-hoc SQL.
AI action safety	100% of AI write attempts routed through Action Catalog with audit logs.
Data freshness	Critical operational marts within agreed SLA; alerts on breach.

14.3 Exit criteria per phase

Phase	Exit criteria
0	RACI approved; source registry started; security risk register created.
1	Canonical Location/ServiceAreaCoverage model available read-only and accepted by Ops.
2	Warehouse raw/staging pipeline running for Antero + ABC Talk + OBL + CMS.
3	ABC Talk contacts/conversations mapped to canonical Contact/Activity/Signal model.
4	Shipment/TTK/Manifest/Dispatch/Trip model reconciled against Antero facts.
5	OBL consumes at least one official metric snapshot from semantic layer.
6	Controlled write-back action live with audit and approval flow.

15. Lampiran

15.1 Source ledger

Source	Used for
Ontology Master v2.0.3 PDF	12-layer ontology context; Dual Domain; Account/Order/Shipment/TTK hierarchy; Trip/Manifest/ShipmentLeg; ServiceAreaCoverage; Signal; TP/OBL concepts.
GitHub: abc-express/antero-platform README/package/src/lib/db.ts	Antero feature scope, stack, direct PostgreSQL server access pattern.
GitHub: abc-express/abc-talk README/package	ABC Talk feature scope, stack, realtime/server-action architecture.
GitHub: abc-express/abc-obl README/package	OBL scorecard purpose, stack, environment separation guidance.
GitHub: abc-express/abc-cms package/config/database.ts	Strapi CMS stack and PostgreSQL configuration.
GitHub: abc-express/abc-ontology README	Reference architecture for bounded contexts, ontology, read-write loops, action runtime.
Supabase project and table inspection	Project mapping, health/region, table inventories, row counts, RLS/security advisories.

15.2 Suggested canonical tables - first release

```
core_locations
core_location_aliases
core_service_area_coverage
core_regional_offices
core_accounts
core_contacts
core_contact_account_links
core_conversations
core_activities
core_signals
core_shipments
core_ttk
core_trips
core_manifests
core_shipment_legs
core_dispatches
core_tp_calculations
core_obl_metric_snapshots
source_entity_mappings
entity_conflicts
survivorship_rules
audit_events
outbox_events
```

15.3 Event taxonomy draft

Event	Producer	Consumer
location.conflict_detected	MDM job	Steward dashboard, notifications.
location.resolved	Steward action API	Warehouse, CMS cache, pricing, order acceptance.
conversation.message_received	ABC Talk	Activity builder, signal engine.
conversation.negative_sentiment_detected	ABC Talk AI/rule engine	Signal engine, AM dashboard.
shipment.status_changed	Antero	Activity builder, SLA metrics, customer

Event	Producer	Consumer
		notification.
manifest.closed	Antero	Warehouse, TP allocation, OBL metrics.
trip.cost_finalized	Antero/Finance integration	TP calculation, finance marts.
obl.metric_snapshot_published	Warehouse semantic layer	OBL app.
signal.resolved	Ontology Core	ABC Talk/Antero/OBL notifications.

15.4 Glossary

Term	Definition
Operational Core	System/layer that controls operational objects and writes with audit. In MVP, centered around Antero.
Analytical SSOT	Warehouse/semantic layer that provides official analytical truth and metrics.
Canonical ID	Stable cross-system identifier for an entity. Source IDs map to it via source_entity_mappings.
Source of Record	System responsible for original transaction in a bounded context.
Source of Truth	Canonical truth for cross-system decision/reporting.
MDM	Master data management: matching, merging, conflict resolution, stewardship, and survivorship.
Action Catalog	Versioned list of allowed read/write actions for apps and AI Agents.
Outbox Event	Reliable event emitted after operational write for downstream consumers.
ShipmentLeg	Junction object between Shipment and Manifest for multi-leg journey and cost/SLA attribution.
ServiceAreaCoverage	Living registry of kab/kota coverage, serving RO/agent, SLA, and status.

15.5 First 10 implementation tickets

#	Ticket	Acceptance criteria
1	Create source_system_registry spec	Antero, ABC Talk, OBL, CMS registered with owner and classification.
2	Define canonical location schema	Reviewed by Ops/Data; includes aliases, codes, validity, source confidence.
3	Extract read-only location sources	Antero city/province/provinsi, CMS areas, ABC Talk location_resolutions available in raw/staging.
4	Build location matching job	Exact and normalized matching with confidence score and conflict queue.
5	Create service area coverage model	coverage_status, serving_ro, serving_agent, SLA, is_3t fields defined.
6	Document Antero POS->Shipment mapping	Field-level mapping and uncertainties logged.
7	Document Manifest/Trip/ShipmentLeg	Trip/Manifest cardinality and leg derivation

#	Ticket	Acceptance criteria
	mapping	validated with Ops.
8	Build warehouse source freshness tests	Freshness checks for priority tables with alerting design.
9	Design audit_events and outbox_events	Schema, retention, idempotency, and replay rules defined.
10	Security exposure inventory	RLS, anon access, service role usage, and functions reviewed read-only.

Final recommendation

Start with canonical Location + ServiceAreaCoverage and warehouse foundation. This provides immediate value, reduces cross-system conflict, and creates the governance muscle needed before tackling higher-risk write-back actions for Shipment, Manifest, Trip, and AI Agent operations.